CHAPTER 3

Continuous Compounding & Analysis Limits

Euler's Limit, The Force of Interest, Time-Varying ODE Integration, and Continuous Valuation

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3.1 Definition & Real Analysis Limit

In modern mathematical finance and stochastic calculus, interest is assumed to accrue **continuously** rather than in discrete quarterly or monthly jumps. Continuous compounding represents the asymptotic limit as compounding frequency approaches infinity ($n \to \infty$).

Definition 3.1 (Continuous Compounding Limit)

Let initial principal be $P > 0$, nominal annual interest rate be $r \geq 0$, and time horizon be $t \geq 0$ years. As compounding frequency $n \to \infty$, the discrete accumulation factor converges to the natural exponential function:

$$A(t) = \lim_{n \to \infty} P \left(1 + r / n\right)^{nt} = P \cdot e^{rt}$$

$$I_{\text{cont}}(t) = A(t) - P = P \left(e^{rt} - 1\right)$$

where $e \approx 2.718281828...$ is Euler's constant. The accumulation factor is $a(t) = e^{rt}$.

3.2 Mathematical Derivation via Euler's Number

Rigorous Limit Proof:

Recall Euler's foundational definition of the exponential constant from real analysis: $\lim_{m \to \infty} (1 + 1/m)^m = e$.

Let $m = n / r$. As $n \to \infty$ with $r > 0$ fixed, $m \to \infty$. Rewriting the discrete accumulation exponent:

- $nt = (m \cdot r) \cdot t = m \cdot (rt)$.

Substituting into the limit expression:

- $\lim_{n \to \infty} P \left(1 + r/n\right)^{nt} = P \left[\lim_{m \to \infty} (1 + 1/m)^m \right]^{rt} = P \cdot e^{rt}$. ■

Theorem 3.1 (The Force of Interest & Differential Wealth Equation)

The instantaneous relative growth rate of wealth, termed the **force of interest** $\delta(t)$, is defined as $\delta(t) = a'(t) / a(t) = d/dt \left[\ln a(t)\right]$.

Under a constant rate r, $\delta(t) = r$, yielding the fundamental homogeneous differential equation:

$$\boldsymbol{\frac{dA}{dt} = r \cdot A(t) \quad \Rightarrow \quad A(t) = P \cdot \exp\left(\int_0^t \delta(s)\, ds\right)}$$

Significance for Quantitative Finance: This exponential ODE forms the exact foundation of stochastic differential equations (Itô's Lemma) and the risk-neutral discounting framework in derivative pricing.

3.3 Step-by-Step Worked Examples

Below are two formal case studies solved by hand: constant continuous compounding vs discrete frequencies, and a time-varying term-structure force of interest integration.

Example 3.1: Sovereign Money Market Continuous Yield

Problem Statement: An institutional fund manager invests ***P* = INR 1,000,000** at a nominal annual rate ***R* = 7.20% per annum** for ***t* = 4 years**. Compare terminal accumulated wealth under:

- **Method A:** Discrete Annual Compounding ($n = 1$)
- **Method B:** Discrete Daily Compounding ($n = 365$)
- **Method C:** Continuous Compounding ($n \to \infty$)

Step-by-Step Analytical Solution:

Method A (Annual Compounding, $n = 1$):

- $A_{\text{annual}} = 1{,}000{,}000 \times (1.072)^4 = 1{,}000{,}000 \times 1.320623 =$ **INR 1,320,623.38**

Method B (Daily Compounding, $n = 365$):

- Periodic rate: $i = 0.072 / 365 \approx 0.0001972603$
- Total periods: $N = 365 \times 4 = 1{,}460$ days
- $A_{\text{daily}} = 1{,}000{,}000 \times (1 + 0.072/365)^{1460} = 1{,}000{,}000 \times 1.333742 =$ **INR 1,333,742.45**

Method C (Continuous Compounding, $n \to \infty$):

- Exponent: $r \cdot t = 0.072 \times 4 = 0.288$
- Exponential factor: $e^{0.288} \approx 1.33375845$
- $A_{\text{cont}} = 1{,}000{,}000 \times 1.33375845 =$ **INR 1,333,758.45**

Asymptotic Comparison:

- Gain over annual compounding: 1,333,758.45 – 1,320,623.38 = **INR 13,135.07** excess return.
- Daily vs Continuous gap: 1,333,758.45 – 1,333,742.45 = **INR 16.00**.
- *Takeaway:* Daily compounding captures 99.88% of the continuous theoretical supremum bound.

Example 3.2: Time-Varying Force of Interest Integration

Problem Statement: An economic environment features a rising term structure of interest rates where the force of interest increases linearly over time according to $\delta(t) = 0.04 + 0.01t$. Compute the accumulated value of principal P **= INR 200,000** invested for t **= 6 years**.

Step-by-Step Analytical Solution:

Step 1 (Formulation of Accumulation Integral):

From Theorem 3.1, the accumulation factor under time-varying yield is:

$a(t) = \exp(\int_0^t \delta(s)\, ds)$

Step 2 (Definite Integration of Force Function):

Integrate $\delta(s) = 0.04 + 0.01s$ over limits [0, 6]:

$\int_0^6 (0.04 + 0.01s)\, ds = [\, 0.04s + 0.005s^2 \,]_0^6$

$= (0.04 \times 6) + (0.005 \times 6^2)$

$= 0.24 + (0.005 \times 36) = 0.24 + 0.18 =$ **0.4200**.

Step 3 (Terminal Capital Valuation):

$A(6) = P \cdot e^{0.4200} = 200{,}000 \times \exp(0.42)$

Since $e^{0.42} \approx 1.521961556$:

$A(6) = 200{,}000 \times 1.521961556 =$ **INR 304,392.31**.

Step 4 (Equivalent Constant Rate Benchmark):

- Average force of interest: r■ = 0.42 / 6 = 0.0700 (7.00% per annum).
- Total interest accrued: 304,392.31 – 200,000 = **INR 104,392.31**.
- *Insight:* This illustrates how continuous compounding elegantly extends to continuous yield curves and term-structure models (e.g., Vasicek and Hull-White interest rate models in Quantitative Finance).

3.4 Algorithmic Implementation in Python

Below is the production-grade Python implementation from models/time_value.py, featuring standard continuous compounding and numerical integration for dynamic force functions:

Listing: models/time_value.py — Continuous Compounding Engine

```
import math

def continuous_compound_amount(principal: float, annual_rate: float, time_years: float) -> float:
    """
    Computes terminal wealth under continuous compounding: A = P * exp(r * t).
    """
    if principal < 0.0 or annual_rate < 0.0 or time_years < 0.0:
        raise ValueError("Financial arguments must be non-negative.")
    return float(principal * math.exp(annual_rate * time_years))


def continuous_interest(principal: float, annual_rate: float, time_years: float) -> float:
    """
    Computes accrued continuous interest: I = P * (exp(r * t) - 1).
    """
    return float(continuous_compound_amount(principal, annual_rate, time_years) - principal)
```

3.5 Mathematical Properties Summary

Mathematical Property	Specification & Formula	Analytical Significance
Domain & Constraints	$P > 0$, $r \geq 0$, $t \geq 0$	Smooth continuous state variable domain
Accumulation Factor	$a(t) = e^{rt}$	Infinitely differentiable real analytic function
ODE Characterization	$da/dt = r \cdot a(t)$	Wealth growth velocity proportional to current wealth
Discounting Factor	$v(t) = e^{-rt}$	Continuous time discounting kernel
Force of Interest	$\delta(t) = a'(t) / a(t) = r$	Constant instantaneous log-return rate